Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/12/2026
PI Name:	Hein, Travis Ph.D.		
Project Title:	Effects of Microgravity on Ocular Vascular Hydrodynamics		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77807	**Congressional District:**	17
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	12/21/2018	**End Date:**	07/14/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Element changed to Space Radiation (SR). Per NASA-JSC, the Space Radiation Element has taken responsibility for the final augmendation of this investigation. The task was originally under the Human Health Countermeasures Element (HHC) (Ed., 6/20/25). NOTE: End date changed to 07/14/2026 per M. Kirby/JSC (Ed., 6/18/25). NOTE: End date changed to 07/14/2025 per NSSC information (Ed., 7/23/24).		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Loerch, Linda (NASA Johnson Space Center) Tharakan, Binu (Scott & White Memorial Hospital) Macias, Brandon (NASA Johnson Space Center) Lee, Stuart (Wyle Laboratories, Inc./NASA Johnson Space Center) Zawieja, David (Texas A&M University System) Bagher, Pooneh (Texas A&M University System) Cromer, Walter (Texas A&M University System) Glaser, Shannon (Texas A&M University System) Chakraborty, Sanjukta (Texas A&M University System)		

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Task Description:	Spaceflight-associated neuro-ocular syndrome (SANS) is reported to affect ~40% of astronauts completing long-duration spaceflights (as of May 2017) and has been characterized as the development of one or more findings: optic disc edema, hyperopic shifts, globe flattening, cotton-wool spots, or choroidal folds. The leading hypothesis for the development of ocular changes is that prolonged exposure to the headward fluid shift that occurs in weightlessness is the primary instigating factor, and additional factors such as genetic disposition, ambient CO2 on the International Space Station, or on-orbit exercise countermeasures may augment or diminish the development of ocular symptoms. However, the pathophysiology of SANS remains unclear. Evidence for the contribution of intracranial pressure alone in SANS is controversial. Therefore, studies of ocular vascular hydrodynamics are required to clarify if chronic mild elevations of ocular pressure variables compromise ocular structure and function. Since all blood and lymph vessels are compliant, fluid-filled structures whose pressures are strongly influenced by gravity, we propose to focus our studies on the potential changes directly to the ocular vasculature caused by microgravity. Perfusion of the optic nerve and inner retina for sufficient delivery of oxygen and nutrients is dependent on retinal blood flow. The pressure gradient for driving blood flow through the inner retina begins with the arterial pressure in the feed artery, which is the central retinal artery in humans. Changes in retinal blood flow or pressure may contribute to the formation of cotton wool spots and optic disc edema. Optic disc edema, choroidal folds, and optic nerve thickening may also result from ocular venous congestion and/or elevated venous compliance, disruption of the blood-retinal barrier, and/or reduction in ocular lymph flow. There has been no systematic analysis of the ocular vascular changes in microgravity. We have assembled a team of experts in SANS and all 3 main vascular types (arteries, veins, and lymphatics) to address this information gap. Thus, the objective of this application is to determine whether microgravity alters the structure and function of the ocular vasculature at the level of feed arteries, venous exchange and capacitance vessels, and lymph vessels. This provides a novel comprehensive evaluation of the ocular vascular elements. The central hypothesis of this proposal is that microgravity/spaceflight-induced changes in the structure/function of the ocular vasculature lead to alterations in ocular hydrodynamics and promote symptoms of SANS. We will accomplish this objective using in vivo measures of vascular function (retinal artery blood flow, retinal arteriole and venular diameter measurements, and retinal venular permeability measures) and in vitro studies of freshly isolated vascular structure and function (vessel/tissue histology, arterial vasomotor regulation, venous compliance measures, and lymphatic transport characteristics). These studies will be conducted in mice flown in space and the corresponding ground controls to address the following specific aims: 1: Evaluate the effects of microgravity on ocular artery structure/function. 2: Evaluate the effects of microgravity on ocular vein structure/function. 3: Evaluate the effects of microgravity on ocular lymphatic structure/function. Information from these novel studies will provide the first comprehensive analysis of the effects of microgravity on ocular vascular function where the predominant changes associated with SANS in astronauts occur. It will also help define the roles these may play in the etiology of SANS and could lead to the development of countermeasures for SANS. We are also performing a task augmentation to supplement our ocular vascular (artery and lymphatics) findings in mice after spaceflight on the ISS in the RR-23 mission on the ISS. We observed decreased ocular artery contractile function and ocular lymphatic phasic activity in the RR-23 mice, which may result from microgravity and/or ionizing radiation. Therefore, the objective of this additional work is to determine whether ionizing radiation exposure in pigs alters the ocular vasomotor function/gene expression of retinal arterioles and cervical lymphatics. Because we found significant changes in pathology and microRNA/mRNA expression in livers from RR-23 mice, pig liver morphology and gene expression will also be assessed after radiation exposure. The miRNA and mRNA expression will be measured in the retinal arterioles/retinal tissue, cervical lymphatics, and liver tissue from the irradiated pigs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The expected findings from the experiments planned for RR-23 and task augmentation will provide new insight into vascular complications relevant to ocular diseases in humans on Earth, such as glaucoma, diabetic macular edema, ocular hypertension, and radiation retinopathy.
Task Progress:	During the past reporting year, we performed polymerase chain reaction (PCR) array analysis of mesenteric lymph nodes from X-ray irradiated pigs. The array contained 91 transcripts in specific gene pathways related to vascular biology and radiation-related aging/oxidative stress. These pathways include endothelial cell biology, hypoxia signaling pathway, DNA repair and cell senescence, aging, oxidative stress, common cytokines, mitochondria, nitric oxide signaling pathway and VEGF signaling. With the limited sample size (1 control and 1-2 per radiation group), so far, we are not able to perform statistical analysis, but changes are evident in our preliminary results. Our results from the pig mesenteric lymph nodes showed increased mRNA expression of genes for endothelial cell biology, oxidative stress, DNA repair and cell senescence, and inflammation signaling molecules at 1 Gy, 2 Gy, and 4 Gy radiation. The same changes in mRNA expression were observed with pig cervical lymph nodes that we observed in the previous reporting period, but the increases in expression tended to be greater. From all results we have obtained during the first two reporting years, the findings suggest that ionizing radiation alters pathways associated with oxidative stress, cellular senescence, and vascular signaling in the pig retina, liver, and cervical and mesenteric lymph nodes. Future studies are planned to increase the sample size for the pig irradiation groups to allow for sufficient PCR array analysis and insight on the impact of radiation on gene expression in these organs. There has been a delay in performing the second cohort of pig irradiation so this has limited our progress during the past reporting year.
Bibliography Type:	Description: (Last Updated: 07/07/2026)

Abstracts for Journals and Proceedings	Pham DL, Chakraborty S, Glaser SS, Cromer WE, Phan LM, Bagley BD, Rivera F, Rigor RR, Zawieja DC, Hein TW. "Impact of ionizing radiation on gene expression in the pig retina, liver, and lymph nodes." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026
Articles in Peer-reviewed Journals	Daniel Estrella L, Sveeggen TM, de la Guardia G, Cacho J, Stauch KL, Bagher P. "A systematic review of the cerebrovascular adaptations following exposure to spaceflight or ground-based analogs: Lessons from human and animal studies." npj Microgravity. 2025 Dec 12. Review. https://doi.org/10.1038/s41526-025-00540-6 ; PMID: 41387730; PMCID: PMC12765887 , Dec-2025